TALENTLENS 2040: AI-POWERED INTELLIGENT RECRUITMENT AND RESUME SCREENING SYSTEM

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***Abstract*—**

Recruitment is an important process for organizations, but manually reviewing a large number of resumes is time consuming and often leads to inconsistent candidate evaluation. This paper presents TalentLens, an AI powered resume screening and candidate matching platform designed to improve the efficiency and accuracy of the recruitment process. The proposed system allows recruiters to enter job descriptions, select required technical and professional skills, and upload multiple candidate resumes in PDF, DOCX, and TXT formats. The system extracts relevant information from resumes and compares candidate qualifications with the selected job requirements using automated skill matching and optional contextual artificial intelligence. It generates candidate matching scores, identifies relevant skills, highlights missing or unclear qualifications, and provides supporting evidence to assist recruiters in evaluating applicants. The platform also provides an interactive dashboard for viewing and comparing candidate profiles. TalentLens aims to reduce manual screening effort, improve transparency, and support consistent recruitment decisions while maintaining human involvement in the final selection process.

***Keywords* — Artificial Intelligence, Resume Screening, Recruitment Automation, Candidate Matching, Skill Extraction, Natural Language Processing, Human Resource Management, Applicant Tracking System.**

I. INTRODUCTION

Recruitment plays a significant role in the growth and success of every organization. Selecting the right candidate for a particular job position requires careful evaluation of an applicant's skills, qualifications, experience, and technical knowledge. However, traditional recruitment processes involve manually reviewing a large number of resumes, which is time consuming, repetitive, and difficult to manage efficiently. Recruiters may also face challenges in identifying suitable candidates when resumes contain different formats, terminology, and descriptions of similar skills.

With the advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), automated recruitment systems have become an important area of research. These technologies provide opportunities to extract relevant information from resumes, understand job requirements, and compare candidate qualifications with organizational needs. AI-based recruitment solutions can assist human resource professionals by reducing repetitive screening tasks and providing structured information for candidate evaluation.

This paper presents TalentLens, an AI-powered resume screening and intelligent candidate matching platform designed to improve the recruitment process. The proposed system allows recruiters to enter job descriptions, select specific skills required for a particular position, and upload multiple resumes in supported formats such as PDF, DOCX, and TXT. The system extracts relevant information from the uploaded resumes and evaluates candidates based on the selected skills and job requirements.

TalentLens incorporates automated skill identification, resume analysis, candidate matching scores, and evidence-based evaluation. It also supports optional contextual AI matching to identify relevant qualifications even when candidates use different terminology to describe similar skills. The platform provides an interactive dashboard through which recruiters can review candidate profiles, identify matching skills, understand missing or unclear qualifications, and compare applicants.

An important feature of TalentLens is recruiter-controlled skill selection, which allows recruiters to define the specific technical and professional competencies required for each job position. This makes the screening process more flexible and adaptable to different recruitment requirements.

The primary objective of TalentLens is to reduce manual resume screening effort, improve the organization of candidate information, and support more consistent recruitment decisions. The system is designed as a decision-support tool rather than an independent hiring authority,

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ensuring that final candidate selection remains under human supervision.

A. Background and Motivation

The increasing number of job applications has created a need for efficient resume screening methods. Traditional manual screening requires recruiters to examine each resume individually, making the process challenging when organizations receive hundreds or thousands of applications.

TalentLens addresses this challenge by introducing automated resume processing and skill-based candidate matching. The platform aims to make recruitment more efficient while providing recruiters with greater control over candidate evaluation.

B. Problem Statement

Existing manual resume screening methods require considerable time and effort to identify candidates who meet specific job requirements. Differences in resume formats, skill descriptions, and professional terminology make candidate comparison difficult. Furthermore, basic keyword-based screening systems may fail to recognize relevant skills when candidates describe their experience using different words.

Therefore, there is a need for an intelligent resume screening platform that can extract resume information, identify relevant skills, compare candidate qualifications with recruiter-defined requirements, and provide understandable evaluation results.

C. Proposed Approach

TalentLens provides an integrated platform for automated resume screening and candidate evaluation. Recruiters can specify job requirements, select required skills, and upload multiple resumes. The system processes the documents, extracts relevant text, and performs skill-based matching using local keyword analysis or optional AI-assisted contextual evaluation.

The results are presented through an interactive dashboard containing candidate matching scores, identified skills, missing or unclear requirements, and supporting resume evidence. This approach helps recruiters make informed decisions while retaining human control over the recruitment process.

D. Organization of the Paper

The remainder of this paper is organized as follows. Section II reviews existing recruitment approaches and related work. Section III describes the preparation of the proposed system, including the technologies used and the candidate matching methodology. Section IV presents the system design and its functional modules. Section V explains the implementation. Section VI discusses the results and the evaluation of the system. Finally, Section VII concludes the paper and presents possible future enhancements.

II. Related Work

Automated support for candidate selection has been studied from several directions. This section groups the relevant work into four categories and then explains where TalentLens fits.

A. Keyword and Statistical Matching

Most applicant tracking systems rely on lexical matching: a requirement counts as satisfied when its words appear in the resume. Classical information-retrieval techniques refine this idea by weighting terms according to how informative they are, and the best-known scheme is term frequency–inverse document frequency (TF-IDF) [1], [2]. Such methods are fast, deterministic, and easy to explain, which is why they remain in wide use. Their main weakness is vocabulary mismatch. A resume that mentions "containerized deployments" may describe the same competence as a requirement for "Docker," yet a purely lexical comparison treats the two phrases as unrelated.

B. Learning-Based Person–Job Fit

A second line of research treats matching as a learning problem. Zhu et al. proposed a convolutional neural network that maps job postings and resumes into a shared representation and can indicate which individual requirement items a candidate satisfies [3]. Qin et al. used an ability-aware neural network to model the abilities that a job demands and those that a candidate's work history demonstrates [4]. Work on the job side shows that postings are often incomplete: Bhola et al. observed that a large share of postings on a national job portal omit relevant skills, and they trained a BERT-based multi-label model to recover them [5]. Such systems perform well when large collections of labelled applications are available. An organization without that history, or one that changes its criteria from one vacancy to the next, cannot easily reuse them.

C. Pretrained Language Models and Generative AI

The transformer architecture [6] made it practical to pretrain language models such as Bidirectional Encoder Representations from Transformers (BERT) [7], which produce context-dependent representations, and Sentence-BERT, which yields sentence embeddings that can be compared efficiently by similarity [8]. Large generative models can follow instructions given in a prompt without task-specific training [9], and retrieval-augmented generation grounds model output in supplied documents [10]. For resume screening, these developments make it possible to judge whether a passage supports a requirement even when different words are used. They also introduce risks: generated text may be unsupported by the source, and sending candidate documents to an external service raises privacy concerns.

D. Fairness, Transparency, and Regulation

Hiring is a sensitive application area. Bogen and Rieke examined how hiring algorithms can affect equity and bias [11], and a widely reported case involved an experimental recruiting tool at Amazon that was abandoned after it was found to disadvantage women [12]. Raghavan et al. studied the claims and practices of vendors of algorithmic hiring tools with respect to bias [13], and Mehrabi et al. surveyed the sources of bias and the definitions of fairness in machine learning [14]. Regulation is emerging as well. The European Union's Artificial Intelligence Act classifies systems used for recruitment and for evaluating candidates as high-risk and requires human oversight of them [15], and India's Digital Personal Data Protection Act, 2023 sets obligations for processing personal data such as that contained in resumes [16]. Explanation methods such as LIME show how individual predictions can be made open to inspection [17].

E. Positioning of TalentLens

TalentLens does not attempt to replace trained prediction models or to automate hiring. It addresses a practical gap between plain keyword filters and models that need historical

training data. The recruiter defines the criteria for each vacancy; a deterministic keyword mode provides a transparent baseline; an optional contextual mode helps with vocabulary mismatch; and every result is accompanied by evidence taken from the resume so that a person can verify it. The score is advisory and the decision stays with the recruiter, consistent with the oversight principles discussed above [15].

III. Preparation of the Proposed System

Before implementing TalentLens, the project requirements, system architecture, functional modules, and data flow are identified. The development process focuses on building a structured resume screening platform that can process candidate resumes and compare their qualifications with recruiter-defined job requirements.

A. Abbreviations and Acronyms

The following abbreviations are used throughout this paper:

- AI – Artificial Intelligence
- NLP – Natural Language Processing
- API – Application Programming Interface
- PDF – Portable Document Format
- DOCX – Microsoft Word Open XML Document
- UI – User Interface
- HR – Human Resources
- ATS – Applicant Tracking System

Abbreviations are expanded when they first appear in the main text to ensure clarity and readability.

B. System Requirements and Technologies

TalentLens is developed using a combination of frontend and backend technologies to support resume processing and candidate evaluation.

The major technologies used in the implementation include:

- **Frontend:** HTML, CSS, and JavaScript for developing the interactive recruitment dashboard.
- **Backend:** Python and FastAPI for handling resume uploads and processing requests.
- **Document Processing:** PyPDF and Python DOCX libraries for extracting text from PDF and DOCX resumes.
- **AI Integration:** Optional OpenAI API integration for contextual resume and job requirement comparison.
- **Skill Matching:** A keyword and skill-alias matching mechanism for identifying relevant candidate qualifications.
- **Development Environment:** Visual Studio Code for writing, testing, and debugging the application.

The system supports PDF, DOCX, and TXT resume formats. File uploads are subject to size and format restrictions.

C. Candidate Matching Methodology

The proposed system evaluates candidate resumes based on the skills selected by the recruiter and the requirements specified in the job description.

In local keyword matching mode, the system identifies the required skills present in the extracted resume text and calculates a matching score.

The matching score is calculated as follows:

$$S = (M / N) \times 100 \quad (1)$$

where S is the matching score, M is the number of matched required skills, and N is the total number of required skills.

For example, if a recruiter selects eight required skills and a candidate's resume contains five matching skills, the resulting keyword matching score is 62.5%.

When contextual AI mode is enabled, the system additionally evaluates the meaning and context of resume information against the job requirements. The results include identified skills, missing or unclear qualifications, and supporting evidence.

These scores are advisory indicators and do not represent the probability of a candidate's job performance or suitability.

D. Common Implementation Considerations

The following considerations are addressed during the development of TalentLens:

- **Resume Format Compatibility:** The system accepts PDF, DOCX, and TXT documents for text extraction.
- **Recruiter Control:** Recruiters can select required technical and professional skills according to each job position.
- **Candidate Evaluation:** Resumes are compared against the selected job requirements.
- **Evidence-Based Results:** The system provides supporting resume excerpts to help recruiters verify identified qualifications.
- **Data Privacy:** Resume information must be handled securely, and external AI processing requires appropriate authorization.
- **Human Supervision:** Final recruitment decisions remain with human recruiters rather than being made automatically by the system.

The proposed platform aims to provide an efficient, flexible, and transparent approach to resume screening and candidate comparison.

IV. System Design and Presentation

The TalentLens platform is organized into multiple functional modules to simplify resume screening and candidate evaluation. The system provides an interactive interface for recruiters to enter job requirements, select required skills, upload candidate resumes, and review matching results. The following subsections describe the organization and presentation of the proposed system.

A. System Architecture

Fig. 1 shows the overall architecture. The recruiter works in a browser-based dashboard that sends the job description, the selected skills, and the uploaded resumes to a backend service. The backend validates each file, extracts its text, and passes the text to the skill-matching engine. When the recruiter enables contextual mode, the engine also sends the job requirements and the resume text to an external AI service and merges the structured answer with the local result. A scoring component then produces a record for each candidate, and the dashboard presents these records to the recruiter. Keeping extraction, matching, and scoring separate allows each part to be tested and replaced independently, for example by substituting a different document parser or AI provider.

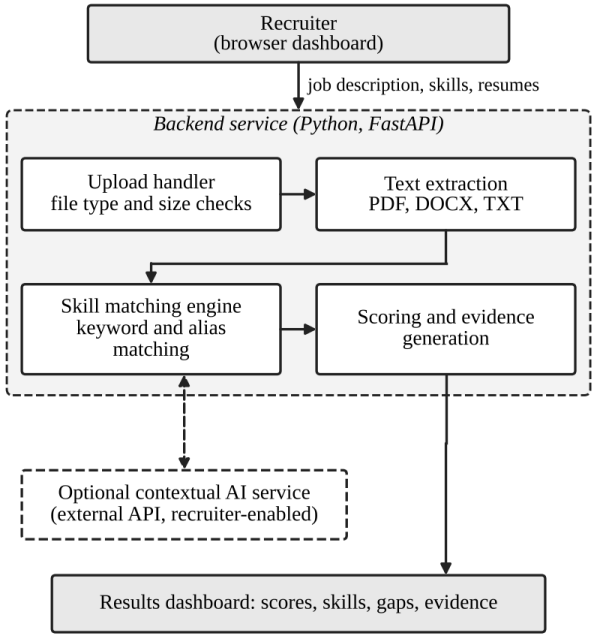


Fig. 1. System architecture of TalentLens.

B. Identification of System Modules

TalentLens consists of several interconnected modules that perform different stages of the recruitment process. TABLE I summarizes them.

1) Recruiter Interface: Provides a dashboard where recruiters can enter job descriptions, define job roles, and manage resume screening activities.

2) Skill Selection Module: Allows recruiters to select required technical and professional skills from suggested options or enter custom skills. These selected skills become the primary matching criteria.

3) Resume Upload Module: Supports uploading multiple candidate resumes in PDF, DOCX, and TXT formats.

4) Resume Text Extraction Module: Extracts readable text and relevant information from uploaded documents using Python-based document processing libraries.

5) Candidate Matching Module: Compares extracted resume information against recruiter-selected skills and job requirements using keyword matching or optional contextual AI analysis.

6) Candidate Evaluation Module: Generates matching scores, identifies matched skills, highlights missing or unclear requirements, and presents supporting resume evidence.

7) Results Dashboard: Displays candidate profiles and screening results in an organized interface to support recruiter review and comparison.

TABLE I. TALENTLENS FUNCTIONAL MODULES

Module	Function
Recruiter Interface	Enter job requirements
Skill Selection	Define required skills
Resume Upload	Upload candidate documents
Text Extraction	Extract resume information
Candidate Matching	Compare skills and requirements
Candidate Evaluation	Generate scores and evidence
Results Dashboard	Display candidate analysis

C. Workflow

A screening session follows the sequence in Fig. 2. The recruiter first enters the job description and selects the required skills, either from suggested options or by typing custom ones. Resumes are then uploaded in a batch. Each file is checked for type and size, and a file that fails the check is reported without interrupting the rest of the batch. Text is extracted from the accepted files and matched against the selected skills. The recruiter chooses the matching mode: local keyword matching, or contextual AI matching when external processing is authorized. Finally, the system assembles a record for every candidate and displays the batch on the dashboard, where the recruiter reviews the evidence and makes the decision.

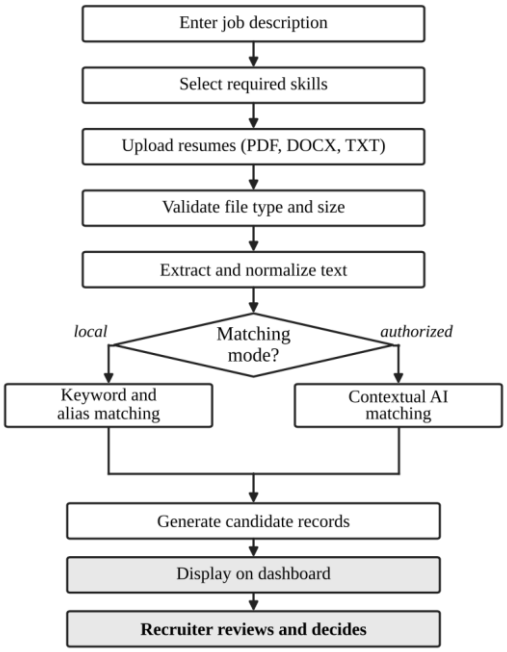


Fig. 2. Workflow of a TalentLens screening session.

D. Candidate Evaluation Output

Each processed resume yields an evaluation record whose fields are listed in TABLE II. The fields are chosen so that every number shown on the dashboard can be traced back to text in the resume.

TABLE II. FIELDS OF THE CANDIDATE EVALUATION RECORD

Field	Description
Matching score	Percentage of required skills found, as defined in (1)
Matched skills	Required skills that have support in the resume
Missing skills	Required skills for which no support was found
Unclear skills	Skills mentioned only indirectly, flagged for recruiter review
Evidence excerpts	Resume passages that support each matched skill
Matching mode	Keyword or contextual mode that produced the record
Status	Recruiter's review state for the candidate

The figures and tables provide a structured explanation of the system and help readers understand how the individual modules work together.

V. IMPLEMENTATION

TalentLens is implemented as a web application with a Python backend and a browser frontend. This section describes the main components and the decisions behind them.

A. Backend Service and Document Processing

The backend is written in Python with FastAPI [18], which exposes the upload and screening operations as web endpoints and validates incoming data through type declarations. Uploaded files are checked against the permitted formats (PDF, DOCX, and TXT) and a maximum size before any processing starts. Text from PDF files is read with the pypdf library [19], text from DOCX files with python-docx [20], and TXT files are read directly. Because PDF is a layout-oriented format, extracted text can contain broken lines or merged words, so the text is normalized by collapsing whitespace and folding case before matching. A scanned PDF that consists only of images yields no text and would require optical character recognition (OCR), which is outside the current scope.

B. Skill Matching Engine

Skill matching is built on a skill dictionary in which each skill has a canonical name and a set of aliases, such as "JS" and "JavaScript." During matching, each alias is searched for in the normalized text. Two details matter for accuracy. First, matches must respect word boundaries, so that a short skill such as "Java" is not reported when the text contains only "JavaScript." Second, a skill is counted once no matter how many aliases or repetitions appear, so that the score in (1) is not inflated by repetition. Recruiters can also add custom skills, which are treated as extra dictionary entries for that screening session. Algorithm 1 summarizes the local matching procedure.

Algorithm 1. Local keyword matching

Input: resume text T; required skills R, each with a set of aliases
Output: score S; matched set M; missing set U; evidence E

```
T ← normalize(T);  M ← { };  E ← { }
for each skill r in R do
    for each alias a of r do
        if a occurs in T at word boundaries then
            add r to M;  E[r] ← sentence containing a;  break
U ← R − M
S ← 100 × |M| / |R|
return S, M, U, E
```

C. Contextual AI Mode

When enabled, contextual mode sends the job requirements, the selected skills, and the extracted resume text to a large language model through the OpenAI API [21]. The prompt asks the model to judge each required skill as supported, unclear, or absent, and to quote the resume passage that justifies its judgment, with the answer returned in a fixed structured format (JavaScript Object Notation, JSON). Model output is treated as untrusted input: the response is parsed against the expected structure, and a response that cannot be parsed is discarded in favor of the local result. Because the evidence passages are intended to be verbatim, they can be compared with the extracted text. The mode is optional because it transmits candidate data to an external provider. It should be used only when the organization has the authority to do so and a lawful basis for processing the data [16]. The API key is kept on the server and is never exposed to the browser.

D. Recruiter Dashboard

The frontend is implemented with HTML, CSS, and JavaScript. After a batch has been processed, the dashboard lists candidates ranked by matching score and lets the recruiter open a detailed view for each of them. The detailed view shows the score, the matched and missing skills, and the evidence passages. A comparison view places selected candidates side by side. Ranking is only a convenience for ordering the review. The evidence is displayed next to every score so that a lower-ranked candidate with relevant experience is not overlooked.

E. Privacy and Security Considerations

Resumes contain personal data, so the design follows the principle of data minimization: only the text needed for matching is processed, and external processing is optional. Restrictions on file type and size limit the risk of malformed or oversized uploads. Attributes that should not influence a decision, such as name, gender, age, or photograph, are not used by the matching engine, which compares resumes only against the skills the recruiter has selected.

VI. RESULTS AND DISCUSSION

This section describes what TalentLens produces, illustrates its scoring with a worked example, compares its two matching modes, and defines the protocol for quantitative validation. The example uses hypothetical resumes constructed for illustration. It is not an experimental result, and no accuracy claims are made from it.

A. Functional Results

The prototype supports the complete workflow from job entry to candidate review. For a batch of uploaded resumes it returns, for every candidate, a matching score, the lists of matched and missing skills, and supporting excerpts, and it presents the batch on the dashboard in ranked order. Because every figure is derived from the skills the recruiter selected, running the screening again with a different selection produces a different ranking, which lets a recruiter test how sensitive the order is to each requirement.

B. Worked Example

Consider a vacancy with eight required skills: Python, FastAPI, SQL, REST APIs, Git, Docker, machine learning, and communication. TABLE III lists three hypothetical

candidates scored with (1). Candidate B reproduces the five-of-eight case given in Section III-C, where 5/8 × 100 = 62.5%.

TABLE III. WORKED EXAMPLE OF KEYWORD MATCHING (HYPOTHETICAL CANDIDATES)

Candidate	Matched	Unmatched skills	Score (%)
A	7/8	Docker	87.5
B	5/8	SQL, Docker, communication	62.5
C	3/8	FastAPI, SQL, Docker, machine learning, communication	37.5

Suppose that Candidate C's resume states that the candidate "built automated pipelines for training and evaluating prediction models" but never uses the phrase "machine learning." Keyword mode reports the skill as missing. Contextual mode could instead mark it as unclear or supported and quote the passage, so that the recruiter can judge it. This is the situation the optional mode is designed for. The example also shows why a score should not be read as a verdict: it measures overlap with a list of skills chosen by the recruiter, not the ability of the candidate.

C. Comparison of the Two Matching Modes

TABLE IV compares the modes. The comparison rests on the design properties of each mode and not on measured performance. The two modes are complementary. Keyword mode gives a stable baseline and needs no external service, while contextual mode can widen coverage at the price of run-to-run variability and the sharing of candidate data.

TABLE IV. COMPARISON OF MATCHING MODES

Property	Keyword mode	Contextual AI mode
Different wording	Only listed aliases	Matched by meaning, not guaranteed
Repeatability	Deterministic	May vary between runs
Evidence	Matched text shown	Quoted passage, to be verified
Data leaves system	No	Yes, to external service
Cost per resume	None	API usage cost

D. Validation Protocol

Quantitative evaluation is required before the scores can be relied on in practice. The proposed protocol uses a set of resumes for one or more vacancies in which experienced recruiters independently label each resume as suitable or not suitable and rank the suitable ones. Precision at k [2] is the fraction of the top k resumes returned by the system that the recruiters labelled suitable:

$$P@k = |Top_k \cap Relevant| / k \quad (2)$$

Agreement between the system ranking and the recruiter ranking is measured with Spearman's rank correlation coefficient [22]:

$$\rho = 1 - 6 \Sigma d_i^2 / (n(n^2 - 1)) \quad (3)$$

where d_i is the difference between the two ranks assigned to resume i and n is the number of ranked resumes. The extraction success rate, which is the share of uploaded files from which usable text is obtained and is reported for each format, and the processing time per resume describe practical performance. Keyword and contextual modes should be compared on the same set, and the contextual evidence should be audited by checking whether each quoted passage actually appears in the resume. Including resumes written in different styles also reveals whether the skill vocabulary disadvantages candidates who describe the same skills in different words.

E. Limitations and Ethical Considerations

Several limitations apply. The score measures overlap with the selected skills and says nothing about the quality of experience, seniority, or potential. Coverage depends on the skill dictionary and its aliases, so missing entries produce false negatives. Text extraction from complex layouts, tables, or scanned documents can be incomplete. Contextual output can be wrong or unsupported even when it reads fluently, which is why its evidence must be verified. The prototype has not yet been validated against recruiter judgments, as described in Section VI-D. Finally, skill vocabularies can reflect the background of the people who wrote them, so recruiters should review the selected skills and aliases from time to time for unintended exclusion. For these reasons TalentLens is positioned as a decision-support tool, and final decisions remain with human recruiters.

VII. CONCLUSION AND FUTURE WORK

This paper presented TalentLens, a resume screening and candidate matching platform that combines recruiter-defined skill criteria, transparent keyword matching, and optional contextual AI analysis. The main design choices are recruiter-controlled skill selection, resume evidence for every score, an optional external AI step, and a final decision that stays with a human. Together they aim to reduce manual screening effort while keeping the process open to inspection.

Several extensions are planned. The first is to carry out the validation protocol of Section VI-D on resumes labelled by recruiters. The second is to supplement dictionary matching with embedding-based similarity [8], which could widen coverage without sending data to an external service. Further work includes OCR support for scanned resumes, audits that compare score distributions across candidate groups for possible bias [14], integration with existing applicant tracking systems, and support for resumes in multiple languages.

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